# IPL ANALYSIS

## Power BI Dashboard Report

2008 - 2025 | 18 Seasons | 10 Teams | 1 Billion Fans

Indian Premier League | Ball-by-Ball Analysis | Power BI + DAX

| | | |
|---|---|---|
| **8,671**<br>V Kohli - All-time Runs | **359**<br>CH Gayle - Sixes | **153**<br>MI - Most Wins |
| **229**<br>YS Chahal - Wickets | **287**<br>Highest Team Score | **18**<br>Seasons Covered |

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# Project Overview

This Power BI dashboard provides a comprehensive analysis of the Indian Premier League (IPL) from its inaugural season in 2008 through 2025. The project covers 18 seasons of professional T20 cricket, analyzing over 1,000 matches, 500+ players, and 10 franchise teams. The dashboard was built using Power BI Desktop with advanced DAX measures, Power Query transformations, and a custom dark navy theme designed to reflect the IPL brand identity.

## Objectives

→ Analyze batting and bowling performance across all IPL seasons

→ Track team standings, wins, and historical dominance

→ Identify top performers — runs, wickets, sixes, economy rate

→ Build an interactive season filter to explore year-by-year trends

→ Present findings in a visually compelling, professional dashboard

## Data Sources

| File | Description | Key Columns |
|---|---|---|
| ball_by_ball_data.csv | Every ball bowled across all seasons | batter, bowler, runs, wickets |
| ipl_matches_data.csv | Match-level results and metadata | match_id, season, winner, toss |
| players_data.csv | Player profiles and batting/bowling style | player_id, name, bat_style |
| teams_data.csv | Team names, short codes, and logos | team_id, team_name, short_code |

## Tools & Technologies

| Tool | Purpose |
|---|---|
| Power BI Desktop | Dashboard design, visual creation, report publishing |
| DAX (Data Analysis Expressions) | KPI measures, calculated columns, aggregations |
| Power Query (M Language) | Data cleaning, transformation, table relationships |
| Data Modeling | Star schema — fact table + 3 dimension tables |
| Custom JSON Theme | Dark navy IPL-branded color scheme |

# Dashboard Pages

## Home Page

### Home Page

- IPL logo, brand header — IPL Analysis 2008-2025
- Season selector dropdown — filter entire dashboard by year
- Champion card with team logo — Royal Challengers Bangalore (2025)
- Runner-Up card with team logo — Punjab Kings (2025)
- Navigation buttons — Insight and Statistics pages
- Full-page IPL jersey banner and players image on the right

## Insight Page

### Insight Page

- Season KPI — IPL Season year display (e.g. 2025)
- Champion & Runner-Up cards with team logos per season
- Key stats row — Total 6s, 4s, Matches, Teams, Centuries, Half Centuries, Venues
- Player cards — Total Fours leader, Total Sixes leader, Orange Cap, Purple Cap
- Points Table — full standings sorted by total points per season
- Official links — iplt20.com, Twitter, Facebook, Instagram, ESPN Cricinfo, CricBuzz, YouTube

## Statistics Page

### Statistics Page

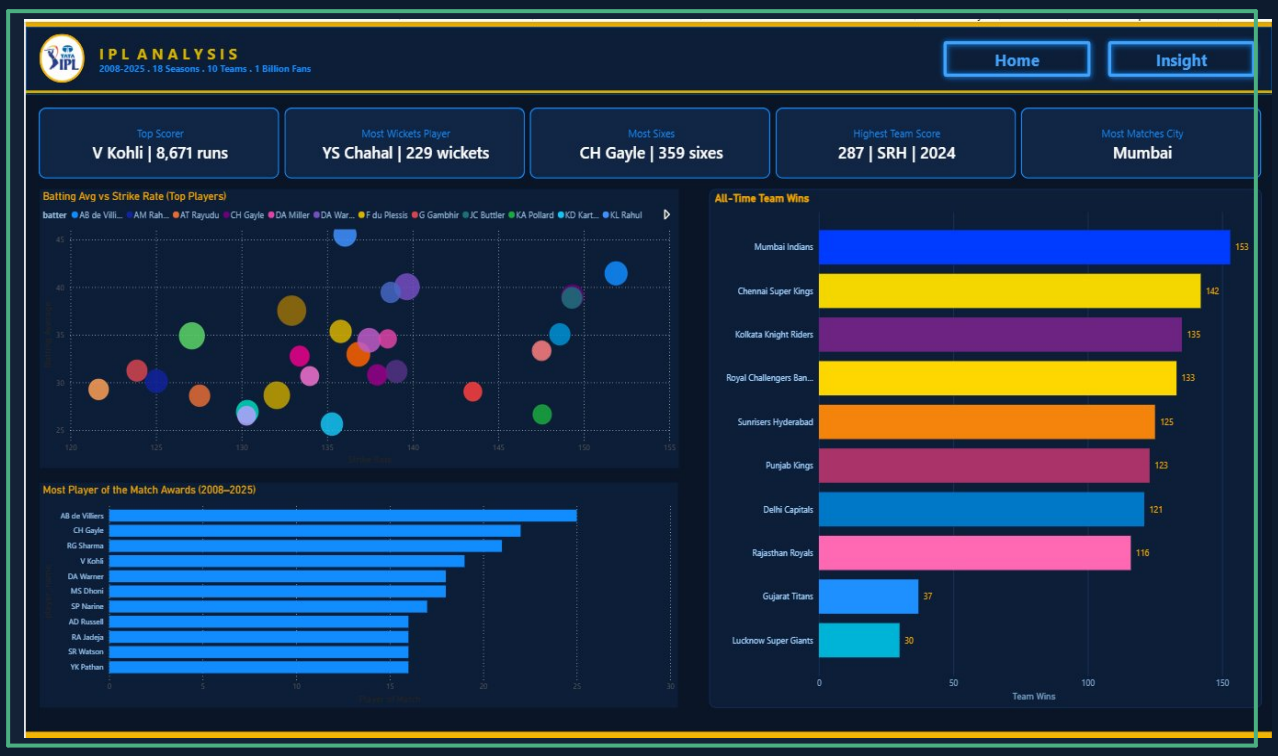
- KPI Cards — Top Scorer, Most Wickets, Most Sixes, Highest Team Score, Most Matches City
- Batting Avg vs Strike Rate — bubble scatter chart (top players with 3000+ runs)
- All-Time Team Wins — horizontal bar chart per franchise
- Most Player of the Match Awards (2008-2025) — top 12 players bar chart

# Dashboard Screenshots

## Home Page

## Insight Page

## Statistics Page



| | | |
|---|---|---|
| **8,671**<br>V Kohli<br>All-Time Runs | **229**<br>YS Chahal<br>All-Time Wickets | **359**<br>CH Gayle<br>Most Sixes |
| **153**<br>Mumbai Indians<br>Most Wins | **287**<br>SRH vs RCB 2024<br>Highest Score | **65%**<br>Teams Field<br>After Toss Win |

## Top Insights

### Batting Dominance

V Kohli leads all-time with 8,671 runs — nearly 1,600 more than the second-placed RG Sharma (7,048). Kohli has been remarkably consistent across 17 seasons.

### Bowling Supremacy

YS Chahal tops the wicket charts with 229 wickets, followed closely by B Kumar (213) and SP Narine (212). Leg-spinners dominate the top of the charts.

### Six-Hitting Power

CH Gayle is the undisputed six-hitting king with 359 sixes — 56 more than RG Sharma in second place. Gayle averaged nearly 1 six per over faced.

### Team Dominance

Mumbai Indians are the most successful franchise with 153 wins across 18 seasons, followed by Chennai Super Kings (142) and Kolkata Knight Riders (135).

### Toss Strategy

Teams winning the toss choose to field 65% of the time, suggesting chasing is the preferred strategy in T20 cricket. Toss winners go on to win the match 51% of the time.

### Record Score

Sunrisers Hyderabad set the all-time highest team score of 287 against Royal Challengers Bangalore in 2024, showcasing the explosive batting evolution in modern T20 cricket.

# Technical Implementation

## Data Model

The project uses a Star Schema with ball_by_ball_data as the central fact table, connected to three dimension tables via foreign key relationships:

| Table | Type | Rows (approx) | Joins To |
|---|---|---|---|
| ball_by_ball_data | Fact | ~250,000+ | ipl_matches_data, teams_data |
| ipl_matches_data | Dimension | ~1,000+ | ball_by_ball_data (match_id) |
| players_data | Dimension | ~500+ | ball_by_ball_data (player names) |
| teams_data | Dimension | 10 | ball_by_ball_data (team names) |

## Key DAX Measures

**Top Run Scorer** — ADDCOLUMNS + MAXX to find player with highest total runs

**Most Wickets** — COUNTROWS with is_wicket = TRUE() filter per bowler

**Batting Average** — DIVIDE(Total Runs, Dismissals) per batter context

**Strike Rate** — DIVIDE(Runs, Balls Faced) * 100, excluding wides

**Economy Rate** — DIVIDE(Runs Conceded, Balls Bowled) * 6 per bowler

**Highest Team Score** — SUMMARIZE by match+team+innings, then MAXX

**Toss Win %** — COUNTROWS filtered by toss_decision / total matches

**Points Table** — Won*2 + NR*1 calculated per team per season

## Power Query Transformations

- Removed NULL values from player_out column and replaced with empty string
- Changed is_wicket, is_wide_ball, is_no_ball columns to Boolean data type
- Created IPL Season calculated column combining season year + "IPL" label
- Set image_url column Data Category to Image URL for logo rendering in tables
- Merged teams_data to get short team names for display in KPI cards
- Filtered super over rows (is_super_over = FALSE) from main analysis

# Challenges & Solutions

### COUNT on Boolean columns
Challenge

Problem: COUNT() does not work on TRUE/FALSE columns in DAX

Solution: Replaced COUNT() with COUNTROWS() and used CALCULATE() with explicit filters

### Cross-filtering between visuals
Challenge

Problem: Clicking one chart was filtering all other charts unexpectedly

Solution: Used Edit Interactions to set all visuals to None interaction mode

### Single dot in scatter chart
Challenge

Problem: Scatter chart showed only one bubble instead of one per player

Solution: Added batter column to Details field well to break aggregation per player

### UNICHAR(10) line break
Challenge

Problem: Line breaks in DAX text measures not rendering in Card visuals

Solution: Split into two separate measures and used Multi-row Card visual instead

### Logo images not showing
Challenge

Problem: Team logo URLs showing as text in table visual

Solution: Set Data Category of image_url column to Image URL in Column Tools

### Season slicer affecting all pages
Challenge

Problem: Season dropdown was filtering stats across pages incorrectly

Solution: Used Edit Interactions to isolate slicer effect per page

# Conclusion & Future Scope

This IPL Analysis Dashboard successfully demonstrates the power of Power BI for sports analytics. By combining ball-by-ball data with match metadata and player profiles, the dashboard delivers actionable insights about player performance, team dominance, and season trends across 18 years of IPL cricket. The project showcases practical skills in DAX, Power Query, data modeling, and professional dashboard design.

## What Was Achieved

- ✓ Built 3 fully interactive dashboard pages with season-level filtering
- ✓ Wrote 15+ complex DAX measures for KPIs, averages, and rankings
- ✓ Designed a professional dark navy theme matching IPL brand identity
- ✓ Implemented star schema data model with 4 related tables
- ✓ Created bubble scatter chart, animated bar charts, and points table
- ✓ Resolved real-world Power BI challenges with Boolean columns and cross-filtering

## Future Improvements

- → Add player comparison tool — side-by-side stats for any two players
- → Build match prediction model using win probability by run rate
- → Add venue analysis — best batting/bowling grounds across India
- → Include partnership analysis — best batting pairs all-time
- → Connect to live API for real-time 2025 season data updates
- → Publish to Power BI Service for web sharing and mobile access

## Author & Contact

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| Project by | Tools | Data | Published |
|---|---|---|---|
| **Nikhil Garg** | **Power BI + DAX** | **IPL 2008-2025** | **April 2026** |